

TAMIL NADU STATE BOARD - STANDARD 12 MATHEMATICS VOLUME 2

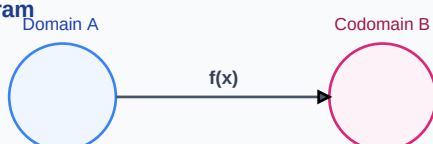
CHAPTER 7: APPLICATIONS OF DIFFERENTIAL CALCULUS

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 12 English Medium
Subject:	Mathematics Volume 2
Chapter Name:	Applications of Differential Calculus

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Function Mapping Diagram

Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Mean Value Theorem	If a function is continuous on $[a,b]$ and differentiable on (a,b) , there exists a point c where tangent equals chord slope.
Point of Inflection	A coordinate point on a curve where the concavity changes from concave upwards to concave downwards, or vice versa.
Local Extrema	Points in the domain where a function achieves its maximum or minimum value relative to neighboring values.

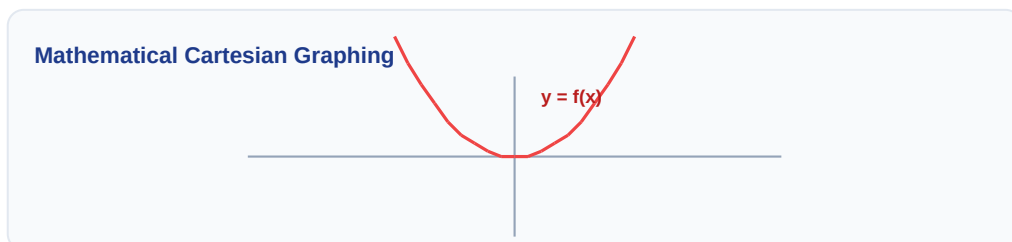
Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

Venn Diagram: $A \cap B$



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:



Essential Formulas, Laws & Identities:

- Rolle's: $f(a) = f(b) \Rightarrow f'(c) = 0$ for c in (a, b)
- Lagrange MVT: $f'(c) = (f(b) - f(a)) / (b - a)$
- Local Maxima: $f'(c) = 0$ and $f''(c) < 0$ | Minima: $f'(c) = 0$ and $f''(c) > 0$
- Inflection Condition: $f''(c) = 0$ and $f'''(c) \neq 0$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Find local extrema of $f(x) = x^3 - 3x + 5$.

Step-by-step Solution:

$$f'(x) = 3x^2 - 3 = 3(x-1)(x+1) = 0 \Rightarrow x = 1, -1.$$

$$f''(x) = 6x.$$

$$\text{At } x = 1: f''(1) = 6 > 0 \Rightarrow \text{Local Minima } (f(1) = 3).$$

$$\text{At } x = -1: f''(-1) = -6 < 0 \Rightarrow \text{Local Maxima } (f(-1) = 7).$$

Linear Algebraic Matrix Representation

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Verify Rolle's Theorem for $f(x) = x^2 - 4x + 3$ on the interval $[1, 3]$.

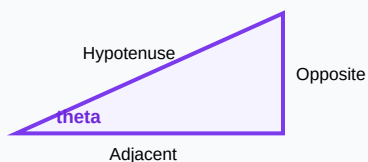
Analytical Solution Strategy:

1) Continuous on $[1, 3]$ (polynomial). 2) Differentiable on $(1, 3)$ with $f'(x) = 2x - 4$.

3) $f(1) = 1 - 4 + 3 = 0$, $f(3) = 9 - 12 + 3 = 0$. $f(1) = f(3)$.

Rolle's condition: $f'(c) = 2c - 4 = 0 \Rightarrow c = 2$. Since 2 lies in $(1, 3)$, Rolle's theorem is verified.

Right-Angled Triangle Geometry



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): State Rolle's Theorem and Lagrange's Mean Value Theorem.

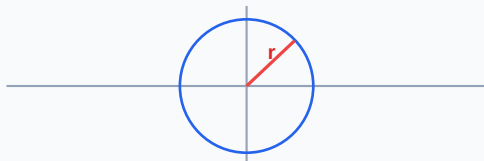
Q2 (2-Mark): Find the intervals of monotonicity for $f(x) = 2x^3 - 9x^2 + 12x$.

Q3 (2-Mark): Determine the concavity of $f(x) = e^x$.

Q4 (5-Mark): Find the point of inflection for $y = x^3$.

Q5 (5-Mark): Find two numbers whose sum is 10 and whose product is maximum.

Circle with Radius Vector on Plane



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Financial Economics

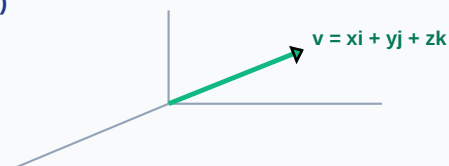
Profit margins are maximized by differentiating revenue equations to locate maximum peaks.

Application Case: Aerospace Layouts

Aircraft skin curvatures are mapped using inflection points to ensure smooth airflows.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

3D Vector Projections (x, y, z)



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Probability Tree Outcome Diagram

